Supplementary Methods

Ability Concealment Does Not Uniquely Reveal the Environment: Learning and Social Opportunity in Gifted Learners

This supplement specifies the five finite-learning experiments reported in the article. It separates environmental reward, sampled information, learned value, and full-information benchmarks. E01–E05 are the complete numerical evidence boundary. The models are stylized computational environments rather than calibrated models of children, and model steps have no calendar-time interpretation.

S1. Common learner and simulation conventions

S1.1 Actions, values, and reward transformation

The common stateless learner assigned one value to each represented action, initialized at

$$Q_0(x) = q_0 = 0.$$

The reference learning rate was $\alpha = .1$, softmax temperature was $\tau = .25$, and negative-reward weighting was $\lambda = 1$. For realized environmental reward R,

$$v_\lambda(R) = \begin{cases} R, & R \geq 0, \\ \lambda R, & R < 0. \end{cases}$$

The transform affected the learning update only; it did not alter the environment's objective payoff. All headline configurations used $\lambda = 1$, so $v_1(R) = R$. For a selected stateless action x_t,

$$Q_{t+1}(x_t) = Q_t(x_t) + \alpha\{v_\lambda(R_t) - Q_t(x_t)\},$$

and unselected values were unchanged. M2 used the state-action update specified in S4.

S1.2 Choice rule

For an available action set A, actions were sampled from

$$Pr(x_t = x) = \frac{\exp\{(Q_t(x) - m_t)/\tau\}}{\sum_{y \in A} \exp\{(Q_t(y) - m_t)/\tau\}}, m_t = \max_{y \in A} Q_t(y).$$

Subtracting the row maximum made the softmax numerically stable without changing its probabilities. Temperature is a design parameter, not a measured learner trait.

S1.3 Payoffs and terminology

The shared reference payoffs were

Symbol	Meaning in the model	Reference value
b	benefit attached to accepted authenticity or masking	1.00
a	authenticity value	0.60
c	masking cost	0.25
r	rejection penalty	2.00

Model masking is the low-visibility action. Authenticity is the represented higher-visibility action. A context is an option with its own acceptance probability; it is not a literal peer group or physical relocation. Neutral, Accepted, and Stigmatized are model states, not diagnoses.

S1.4 Horizons and final windows

E01, E02, E04, and E05 used 500 simulated learners per cell, $T = 2000$ steps, and a 400-step final window (steps 1600–1999). E03 used 160 learners per cell, $T = 1000$, and a 200-step final window (steps 800–999). In general the final window began at

$$t_{start} = T - \lceil .2T \rceil.$$

Final-window modal classifications used action counts, not final Q ordering. They are operational finite-horizon summaries and do not assert convergence.

S2. E01 — feedback-history experiment

S2.1 M0 reward structure and objective boundary

M0 had masking (M) and authenticity (A). Masking returned

$$R_M = b - c.$$

Authenticity was accepted with probability p, returning $b + a$, and rejected otherwise, returning $a - r$. Its expected reward was

$$E[R_A] = p(b + a) + (1 - p)(a - r) = a + pb - (1 - p)r.$$

The authentic-minus-mask advantage was

$$\Delta_0(p) = a + c - (1 - p)(b + r).$$

Thus,

$$p^* = 1 - \frac{a + c}{b + r} = 1 - \frac{.85}{3} = \frac{43}{60} \approx .716667.$$

At the reference values $R_M = .75$. Authenticity was therefore objectively optimal at both primary receptivity values, $p = .75$ and $p = .85$.

S2.2 Conditioning and estimands

Every learner's step-0 action was forced to authenticity. Learners were stratified by whether this first outcome was accepted or rejected. Subsequent actions occupied steps 1 through $T - 1$.

The primary sampling estimand was

$$\Delta \acute{A}_{1:T-1} = \acute{A}_{1:T-1/rejected} - \acute{A}_{1:T-1/accepted},$$

pooled with equal weight across $p = .75$ and .85 within each master seed. The secondary persistence estimand replaced subsequent action rate with final-window mask rate.

In authenticity-optimal binary cells, normalized objective pseudo-regret is an exact action-count translation rather than independent confirmation:

$$\Delta \overline{Regret}_{norm} = -\frac{T-1}{T} \Delta \acute{A}_{1:T-1}.$$

For $T = 2000$, the factor is $1999/2000$.

S2.3 Information manipulations

The controlled baseline supplied feedback only for the selected action. In the forced-authentic condition, 5% of post-initial choices were replaced by an authentic action using a separate intervention stream. In the counterfactual feedback condition, the learner received the authentic outcome that would have occurred on a masking trial. The environment used the same latent acceptance draw for that counterfactual, while counterfactual reward was excluded from the learner's realized objective return. These manipulations alter information allocation inside the model; they are not human treatment estimates.

S2.4 Design and uncertainty

The standard design used $n = 500$, $T = 2000$, five master seeds 2026081101–2026081105, and $p \in \{.75, .85\}$ for the controlled history contrast. Within each master, the two p-specific effects were equally weighted. The reported 95% interval was

$$\acute{\theta} \pm t_{.975,4} \frac{s_\theta}{\sqrt{5}}, t_{.975,4} = 2.776445.$$

The top-level inferential units were the five master-specific effects, not individual simulated learners.

Secondary archived cells compared $\lambda = .5$ and 2.5 at $p = .75$ under the controlled baseline and at $p = .65$ under the maintained baseline. They were kept secondary because the article does not interpret λ as a human rejection-sensitivity construct. No horizon or learning-rate sweep was executed for E01.

S3. E02 — graded action representation

S3.1 Binary and graded architectures

The binary architecture represented $d \in \{0, 1\}$, where $d = 0$ was masking and $d = 1$ was full authenticity. The graded architecture represented five independently learned arms:

$$d \in \{0, .25, .5, .75, 1\}.$$

Experience at one disclosure level updated only that level. It did not generalize to neighbouring values.

For disclosure level d, acceptance probability was

$$Pr(accept/d) = 1 - (1 - p)d,$$

which yields

$$E[R(d)] = ad - c(1 - d) + b\{1 - (1 - p)d\} - r(1 - p)d$$

and hence

$$E[R(d)] = b - c + d\{a + c - (1 - p)(b + r)\} = b - c + d\,\Delta_0(p).$$

The objective is affine in d. Except at exact indifference, a strict optimum is therefore an endpoint; intermediate disclosure is not a distinct static optimum in this model.

S3.2 Conditioning, outcomes, and pairing

E02 used $p \in \{.65, .75, .85\}$, $n = 500$, $T = 2000$, and master seeds 2026081201–2026081205. The first action was full authenticity, and the main comparison used the rejected-initial-outcome stratum at $p = .75$ and $p = .85$, equally weighted within master.

The maintained outcomes were:

1. post-initial rate of any non-zero disclosure in the graded architecture minus post-initial full-authentic resampling in the binary architecture;
2. graded minus binary final-window full-authentic action rate;
3. graded minus binary probability that $d = 1$ was the unique final-window modal arm; and
4. binary minus graded normalized pseudo-regret.

The unique-modal statistic required the objective endpoint to have a strictly larger final-window count than every other represented arm. The broader full-authentic action-rate statistic did not impose modal uniqueness. In authenticity-optimal cells, graded normalized pseudo-regret is exactly the horizon-average disclosure distance $1 - d$; it is not independent welfare evidence.

Binary and graded learners were reinitialized from the same policy/environment stream within master seed and receptivity. Full latent uniform sequences and child IDs were matched within those pairs. Pairing did not cross receptivity values or master seeds. Student-t intervals used five master-specific paired architecture effects as the top-level units.

The $p = .65$ condition was a maintained control rather than a headline decision input. Three- and nine-level graded architectures were planned but not executed and are not represented as sensitivity evidence.

S4. E03 — persistent-state environment and dynamic oracle

S4.1 State and transition kernel

M2 used states Neutral (N), Accepted (A), and Stigmatized (S), with actions M and A. Reward was realized before the next-state transition. The reference transition/reward kernel was:

Current state	Action/outcome	Reward	Next state
N	mask	$b - c$	N
N	authentic, accepted with p	$b + a$	A
N	authentic, rejected with $1 - p$	$a - r$	S
A	mask	$b - c$	N with δ, else A
A	authentic	$b + a$	N with δ, else A
S	mask	$b_2 - c$	N with η, else S
S	authentic, redeemed with $p/2$	$b + a$	A
S	authentic, not redeemed	$a - r_2$	N with η, else S

The reference values were $b_2 = .5$, $r_2 = 2.5$, and $\delta = .02$. Escape η was applied after the current reward on non-redeemed Stigmatized transitions. Masking in Neutral did not reset or create stigma.

S4.2 Learner update

The selected state-action value followed

$$Q_{t+1}(s_t, x_t) = Q_t(s_t, x_t) + \alpha[v_\lambda(R_t) + \gamma \max_{x'} Q_t(s_{t+1}, x') - Q_t(s_t, x_t)],$$

with $\gamma = .9$. The values of unselected state-action pairs were unchanged. The Neutral-state authentic-minus-mask advantage for continuation value V was

$$D_N(V) = \Delta_0(p) + \gamma\{pV_A + (1-p)V_S - V_N\}.$$

This expression makes explicit why the dynamic boundary can differ from the static M0 boundary.

S4.3 Full-information dynamic comparator

For each state-action pair, the oracle used the untransformed environmental reward and known transition probabilities:

$$Q^V(s,x) = \sum_{s'} P(s'/s,x)\{r(s,x,s') + \gamma V(s')\},$$

$$(TV)(s) = \max_x Q^V(s,x).$$

Value iteration began from $V_0(s) = 0$ and repeatedly set $V_{k+1} = TV_k$ until

$$\max_s |V_{k+1}(s) - V_k| < 10^{-10},$$

with a fail-closed maximum of 100,000 iterations. Objective action ties used a 10^{-10} tolerance. The computed envelope was independently checked by enumerating deterministic policies; the maximum recorded discrepancy was 8.99×10^{-10}. The oracle is infinite-horizon and full-information. It is intentionally not information-matched to the finite learner.

S4.4 Grid, classifier, and stability

The exact grid was

$$p \in \{0, .05, \ldots, 1\}, \eta \in \{0, .05, \ldots, .8\},$$

giving $21 \times 17 = 357$ cells. Each cell used $n = 160$, $T = 1000$, and five master seeds 2026081301–2026081305.

Within each master and cell, final-window Neutral-state action counts were aggregated across learners. Neutral exposure was adequate only if:

1. at least 160 Neutral-state final-window visits were observed in total; and
2. at least 32 learners contributed at least one Neutral-state visit.

Equal aggregate mask/authentic counts or inadequate exposure produced an inconclusive learned action. A cell-master was labelled learned_suboptimal_masking only when exposure was adequate, aggregate Neutral counts favoured masking, and the full-information oracle favoured authenticity in Neutral. Stability required the same non-inconclusive class in at least four of five masters.

The reported $243 / 357$ statistic counts grid cells satisfying that learned–objective disagreement classification in every master. The remaining grid contained 107 agreement cells and seven inconclusive cells. Five of those seven had inadequate Neutral exposure in all five masters. At $p = .25$ and $p = .30$ with $\eta = 0$, only one and two masters, respectively, had adequate exposure and agreed on learner_matches_objective; those cells failed the prespecified four-of-five stability rule and were therefore folded into the inconclusive publication category. The statistic is tested-grid coverage, not a fraction of learners, people, or naturally occurring environments.

S5. E04 — contextual search architectures

S5.1 Environmental structure and indexing

M3 contained K contexts. In the primary condition, one context was receptive with $p_{good} = .85$ and the remaining $K - 1$ contexts had $p_{bad} = .20$. Context indices were zero based in the

maintained implementation. In joint-action representations, action index $2g$ denoted masking in context g, and $2g+1$ denoted authenticity in context g.

Masking returned $b-c$ in every context. Authenticity used that context's acceptance probability. With comparison cost $b_f = 0$ in E04, the distinct expected-reward classes were mask, good-context authenticity, and bad-context authenticity. Increasing K duplicated classes but did not change their values or the full-information maximum. The full-information best opportunity was therefore K-invariant in the primary one-good-context design.

S5.2 Four action representations

Architecture	Choice and update
flat_2k	One softmax over the $2K$ joint context–behaviour arms; only the selected joint arm was updated.
hierarchical	A K-way context softmax followed by a two-way behaviour softmax within the selected context; the selected context value and selected within-context behaviour value were both updated from the same realized reward.
global_mask	One shared mask arm plus K context-specific authentic arms; the selected arm alone was updated.
fixed_known_good	A two-action mask/authentic problem in the known good context; context discovery was removed and the architecture was structurally invariant to requested K.

All architectures used the same reference learner settings. Acceptance uniforms were consumed on every trial in the maintained random-number order.

S5.3 Design, canonicalization, and outcome

The requested design crossed the four architectures, $K \in \{1, 2, 4, 8, 16\}$, fixed versus random-per-learner good-context identity, and one, two, or proportional-25%-good-context regimes. Conditions that were structurally equivalent were canonicalized: 120 requested rows mapped to 55 executable cells, recorded in the equivalence table. Each executable cell used $n = 500$, $T = 2000$, and master seeds 2026081401–2026081405.

For each learner, deterministic NumPy first-index argmax was applied to final-window joint-action counts. The learner indicator equalled one only when that selected indexed arm was authentic in a good context. Non-unique modes remained eligible because a tie was resolved to its first indexed maximum. The primary cell response was the mean of this indicator across learners. This is not the superseded rule under which any tied good-authentic maximum would qualify.

A separate strict diagnostic required a unique modal good-authentic arm. It was not substituted for the prespecified primary estimator. The diagnostic retained the direction of the primary flat and hierarchical contrasts.

The main one-good-context curves were accompanied by fixed/random identity controls, a two-good-context control when feasible, the proportional-25% receptive control, global-mask and fixed-known-good ablations, and complete K curves. Planned sensitivity over p_{good}, p_{bad}, and τ was not executed and is not reported as evidence.

S5.4 Inference

Each architecture contrast or K slope was first reduced within master seed. The reported interval then used the five master-specific estimates in the Student-t construction from S2.4. Learners and K cells were not treated as additional independent top-level replicates.

S6. E05 — learned and environmental boundary estimation

S6.1 Environmental boundary

E05 fixed $K = 4$, one good context at index 0, $p_{good} = .85$, $p_{bad} = .20$, and evaluated the flat eight-action and hierarchical representations separately. Comparison cost b_f was subtracted only from accepted authenticity in the good context. Thus,

$$R_M = .75,$$

$$R_G(b_f) = .85(1.6 - b_f) + .15(-1.4) = 1.15 - .85\,b_f,$$

and

$$R_G(b_f) - R_M = .40 - .85\,b_f.$$

The exact environmental crossover was

$$b_f^* = \frac{.40}{.85} = \frac{8}{17} = .470588\ldots.$$

Bad-context authenticity did not become the competing objective optimum in the tested range.

S6.2 Grid and learned response

The complete cost grid was

$$\{0, .30, .325, .35, .375, .40, .425, .45, 8/17, .475, .50, .525, .55, .575, .60, .625, .65, .80\}.$$

The monotone crossing fit used the 16 points inside the dense range $[.30, .65]$, including the exact $8/17$ anchor. The 0 and $.80$ cells were endpoint diagnostics outside that fit.

At each cost and master seed, the learned response was the fraction of 500 learners whose deterministic final-window first-index argmax selected the single good-context authentic arm. It was an across-learner modal-arm classification fraction, not a binary softmax action probability. The same architecture/master policy-environment stream was reinitialized at every cost, providing common-random-number pairing across b_f within an architecture. Flat and hierarchical architectures used separate random-stream families.

S6.3 PAVA fit and crossing

The five master responses were averaged equally at each cost. A non-increasing, equal-weight pool-adjacent-violators fit was then applied to the pooled dense curve (Best & Chakravarti, 1990). The fitted .5 crossing was defined as follows:

1. if a fitted plateau equalled .5, use the midpoint of that plateau's b_f range;
2. otherwise linearly interpolate between the adjacent fitted values that bracketed .5;
3. accept a crossing only when it lay strictly inside $(.30, .65)$;
4. do not extrapolate.

A response already below .5 at .30 was classified no_crossing_below and was left-censored below the tested range. This was the hierarchical result; it was not interpreted as absence of a crossing over all possible costs.

S6.4 Complete-master-curve bootstrap

The crossing interval used 10,000 bootstrap replicates. Each replicate sampled five complete master curves with replacement, averaged the sampled curves, refitted PAVA, and reinverted the fit. Percentile limits at 2.5% and 97.5% were reported only if at least 95% of replicates yielded valid in-range crossings. All 10,000 flat replicates were valid. Zero hierarchical replicates were valid, so no hierarchical point or interval was estimated. The bootstrap unit was a complete master curve, not a learner or individual cost row.

The cost axis orders environmental parameter values. It is not time, attempts, or a point at which a learner gives up. No binary-softmax indifference identity was used to interpret the learned crossing.

S7. Randomization, seed hierarchy, and inferential units

S7.1 Master seeds and semantic streams

The archived master lists were:

Experiment	Master seeds
E01	2026081101–2026081105
E02	2026081201–2026081205
E03	2026081301–2026081305
E04	2026081401–2026081405
E05	2026081501–2026081505

Semantic cell or family identifiers were hashed with SHA-256. The first four little-endian 32-bit words from the digest were supplied to NumPy SeedSequence together with the master and experiment namespace:

Experiment	Archived construction
E01	SeedSequence([master, 1, words]).spawn(2), separating policy/environment and intervention streams
E02	SeedSequence([master, 2, words]).spawn(1) for the matched binary/graded pair
E03	SeedSequence([master, 3, words]).spawn(1); approved horizon/window sentinels alone shared trajectory-prefix families
E04	SeedSequence([master, 4, words]).spawn(1) for policy/environment and independent SeedSequence([master, 40, words]).spawn(1) for identity
E05	SeedSequence([master, 5, words]).spawn(1), reinitialized across cost within architecture/master; bootstrap SeedSequence([2026081597, 50, architecture code])

Vectorized child draws were generated inside each cell/family stream. Child IDs were not additional top-level inferential replicates. E01 acceptance and rejection strata were disjoint outcome strata, not paired learners. E02 paired architectures only within master and receptivity. E04 reduced K and architecture effects inside master. E05 resampled complete master curves.

S7.2 Inferential scope

Student-t intervals in E01, E02, and E04 used five master-specific estimates (df=4). E05's scalar endpoint effects used the same master-level convention, whereas its crossing used the complete-

curve bootstrap. E03's grid fraction was identical across the five selected masters; this records stability across those streams, not zero model uncertainty. Master lists were fixed constants, not draws from a documented population of seeds.

S8. Secondary estimands and sensitivity analyses

Secondary archived evidence was retained to test interpretation boundaries without promoting it to a headline mechanism.

Experiment	Secondary material	Status and role
E01	forced-authentic 5% and counterfactual-authentic feedback; $\lambda = .5 / 2.5$ cells	Archived. Information-channel effects support the sampling interpretation. Weighting remains learner-side sensitivity, not a human construct.
E02	full-authentic rate, unique-modal endpoint, and disclosure-distance pseudo-regret; $p = .65$ control	Archived. These outcomes prevent interpreting increased non-zero disclosure as improved endpoint discovery.
E03	six sentinels; $n = 320$; $T = 2000 / 5000$; windows .1/.4; $\alpha = .03 / .3$; $\tau = .1 / .6$; $\lambda = .5 / 2.5$; $q_0 = .75$; $\gamma = .5 / .99$; harsh boundary cells	Archived where present. They show classifier dependence on learner and dynamic settings. They do not define a population-wide phase boundary.
E04	all four architectures; fixed/random identity; one, two, and proportional-25% good-context regimes; strict unique-mode diagnostic	Archived baseline ablations. Planned p and temperature sensitivity was not run.
E05	five master curves; endpoint contrast; strict unique-mode diagnostic; flat PAVA/bootstrap; hierarchical censoring	Archived. A longer-horizon crossing was planned but not executed.

E03's harsh labelled boundary used $b_2 = 0$, $r_2 = 8$, $\delta = .3$, $\eta = 0$, $\gamma = .9$, and $p \in \{.70, .75, .80\}$. It demonstrates a bounded case in which dynamic masking can be objectively optimal despite a static M0 authenticity advantage; it is not evidence for a region beyond the tested cells.

S9. Evidence provenance and reproduction

S9.1 Three distinct provenance claims

The evidence package distinguishes:

1. ARTIFACT_VERIFIED: existing outputs byte-match the hashes stored in their archived manifests and evidence records;
2. CURRENT_IMPLEMENTATION_REPRODUCIBLE: a maintained procedure can be run from the present code, where separately authorized;
3. HISTORICAL_BIT_REPRODUCTION_VERIFIED: the exact historical source state and outputs can be reconstructed.

All 33 protected E01–E05 outputs are ARTIFACT_VERIFIED. Fingerprint-exact historical generator-source bundles are retained for E04 and E05. E01–E03 are HISTORICAL_SOURCE_STATE_UNRECOVERED.

Historical generator states underlying some archived E01–E03 artifacts are not fully recoverable. The published claims are therefore governed by archived result artifacts, hashes, configurations, and verification records rather than by asserting bit-identical regeneration from the present implementation. This does not imply that the archived artifacts changed; it limits source-level historical reconstruction.

S9.2 Archived summary-artifact identities

Experiment/artifact	SHA-256
E01 effects	4ce67da8b48e8d770388db9e2c6ebe240b868c0696ba90678cc9940de02684d8
E02 effects	11d29c72508649a174b285957dfec3752edff76a099aa76de9f6fe3a4e515a8a
E03 effects	5a1ee392560e94695dfe0d04ff89caeafdb0b58fa2b70e6def365726d66ab3af
E03 stable cells	91814ab4430f63bc5e46596c61718e1fec6705d355b6cab3d8c18d29c1dfaea7

Experiment/artifact	SHA-256
E03 thresholds	0987aea9f6085f81bad791454eaca03127c6144f172a7cc9f41cdad6c316f9bf
E03 cells	198cb7be9b2a93aa89569d9dd656326f521b7a2ec99543733fd6e779e0508851
E04 effects	dd426992f0a1a0ae05542756dc6dca918fbf56a589347676a93a309d08a41d44
E05 analytic	e1045d0593709561761ec8c9ba0684ef1a80e26d6b0b15f12f57434b74f22a42
E05 transitions	96b8d8dc1cf27e3a226282e4d8755effde74e855d81b8f8bbc6db6515c607097
E05 effects	33eb752c89d87ce79ef13bfe196623fb4d4f53c4fff63e03b84af413a68200d5

The five standard manifests under results/manifests retain the normalized configuration, exact parameters, seed hierarchy, environment versions, output paths and hashes, timestamps, and source fingerprints. The recorded execution environment was Python 3.14.6 on macOS arm64 with NumPy 2.4.6; E01, E02, E04, and E05 also recorded Matplotlib 3.10.8.

S9.3 Validation and presentation reproduction

Read-only artifact validation loads the archived manifests and evidence matrix and rehashes their declared outputs. It does not run a simulation. Figure presentation can be regenerated deterministically from the pinned publication-source tables, but that operation is distinct from experiment reproduction. The figure-generation script imports no experiment module or stochastic entry point.

The figure-source hashes used by the final display package are recorded in the accompanying figure-provenance record. Examples include e01_primary_effects.csv (9b9227f43146b96ae7000f572d2249ae6bd4d39bf414715bdd780203afa38d36), e03_grid_map.csv (c5def710116248b60ffcc347fa408519ce5d9094547e486fc73f2388e5babbab), e04_k_response.csv (c01c04359b30d425d518566190dcfb27b10f9003e4123d9a6f05ce44382c11b7), and

e05_crossings.csv

(866697afb0cdd66f6e6d008cc6b06e52786d1e27f7ba3f71601a56512349076b).

No claim in this supplement requires a current-code experiment rerun, and no historical-bit-reproduction claim is made for E01–E03.

S10. Supplementary figure and table inventory

The following inventory lists supplementary material available from the archived configurations and result artifacts. Secondary evidence remains separate from the main Results.

Candidate	Content	Status
Table S1	common parameters, horizons, and master seeds	Available from archived configurations
Figure/Table S2	E01 master-level sampling and information-manipulation diagnostics	Available from archived artifacts
Figure/Table S3	E02 secondary full-endpoint, unique-modal, and regret outcomes	Available from archived artifacts
Figure/Table S4	E03 five-master stability, seven inconclusive cells, sentinels, and threshold brackets	Available from archived artifacts
Figure/Table S5	E04 complete K curves, identity/good-count ablations, equivalence mapping, and strict tie diagnostic	Available from archived artifacts
Figure/Table S6	E05 five master curves, pooled PAVA fit, bootstrap validity, endpoint and strict-mode diagnostics	Available from archived artifacts
Former cross-estimand Figure 6	synthesis across noncommensurate estimands	Not included; comparability review required

The former cross-estimand Figure 6 is not included because its quantities require a separate scientific comparability and interpretation review.

Supplement reference

Best, M. J., & Chakravarti, N. (1990). Active set algorithms for isotonic regression: A unifying framework. *Mathematical Programming*, *47*(1–3), 425–439. https://doi.org/10.1007/BF01580873